

Task List for Python and ML

Complete the following Python exercises in order

1. Python Tasks

a. Basic Arithmetic Operations

- i. Write a program to perform addition, subtraction, multiplication, and division of two numbers provided by the user.

b. Palindrome Checker

- i. Write a function to check if a given string is a palindrome.

c. File Handling

- i. Create a script to read a text file, count the number of words, and write the results to a new file.

d. Data Structures: Lists and Dictionaries

- i. Write a program that stores student data (name, age, grades) in a dictionary and performs CRUD operations.

e. Basic OOP (Object-Oriented Programming)

- i. Create a Student class with attributes for name, age, and grades. Include methods to display details and calculate the average grade.

f. API Request

- i. Use the requests library to fetch data from a public API (e.g., JSONPlaceholder, Weather API, Quotes API). Parse and display specific information.

2. API Development Task Using Flask

a. Objective:

- i. Implement a RESTful API using Flask and create a frontend interface to interact with the API.

b. Requirements:

- i. Implement CRUD operations:
 1. Create (POST), Retrieve (GET), Update (PUT), Delete (DELETE).
- ii. Design a topic for your API. Examples:
 1. A task management system (e.g., To-do List).
 2. A movie review database.

3. A library system to manage books.
 4. You can choose any other topic as well.
- iii. Build a frontend to interact with the API using HTML/CSS/JavaScript or connect Frontend directly in Flask.

3. Machine Learning Project

a. Objective:

- i. Develop a Regression Model using any regression algorithm (e.g., Linear Regression, Decision Tree Regressor).

b. Steps:

- i. Select a dataset of your choice (e.g., housing prices, student grades, or car sales data).
- ii. Train a regression model on the dataset and evaluate its performance.
- iii. Export the trained model into Flask.
- iv. Create a simple Flask API that serves predictions using the model.
- v. Design a frontend to take input for predictions and display the results.

Submission Details

- Deadline: 8th January
- Submission Process:
 - Push your completed tasks to a GitHub repository.
 - Share the repository link via a Google Form that will be provided for submission.

Evaluation will be based on code functionality, clarity, and adherence to the task requirements.